

COURSE TITLE : SECURITY PRINTING
COURSE CODE : 5105
COURSE CATEGORY : E
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 44
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Introduction to security printing	14
2	Security inks and substrates	10
3	Applications of Security printing	12
4	Holograms, Credit Cards and passports	8
TOTAL		44

Course Outcomes :

G .O.	ON THE COMPLETION OF THE STUDY OF THIS MODULE STUDENT WILL BE ABLE :
1	To comprehend about security printing.
2	To understand about Security inks and substrates.
3	To understand currency printing, cheque printing ,numbering and bar-coding.
4	To know about Holograms, Credit Cards and passports security features.

MODULE I INTRODUCTION TO SECURITY PRINTING

1.1.0. To comprehend about security printing.

- 1.1.1. To define security printing.
- 1.1.2. To understand goals of security printing.
- 1.1.3. To list various security features.
- 1.1.4. To describe various printing methods used for security printing.

MODULE II SECURITY INKS AND SUBSTRATES.

2.1.0 To understand about Security inks and substrates.

- 2.1.1 To know the basic principles of Security inks and substrates
- 2.1.2 To list different types of Security inks and substrates
- 2.1.3 To list different security threads.
- 2.1.4 To discuss about water mark.

MODULE III APPLICATIONS OF SECURITY PRINTING

3.1.0 To understand currency printing, cheque printing ,numbering and bar-coding.

- 3.1.1 To state currency printing.
- 3.1.2 To list various security features in currency.
- 3.1.3 To discuss cheque printing.
- 3.1.4 To define barcode
- 3.1.5 To identify various barcodes.
- 3.1.6 To discuss about QR code.

MODULE IV HOLOGRAMS, CREDIT CARDS AND PASSPORTS

4.1.0 To know about Holograms, Credit Cards and passports security features.

- 4.1.1 To compare different Holograms
- 4.1.2 To define hologram
- 4.1.3 To list credit card security features.

CONTENT DETAILS

MODULE I

Security printing –Definition. Goals of security printing. Security features- water mark, security thread, latent image, micro lettering, Intaglio Printing, identification mark, see through register, hologram etc.

Basic principles of security design features - Pantograph screens, Void pantograph Screen, ODT - optical deterrent technology, Guilloches, Warning bands, Code safe, High resolution graphics and Padlock icon .

Various printing methods used for security printing.

MODULE II

Basic principles of security inks – Trademark colors, Color changing ink, Magnetic ink, Copy protection ink, Erasable ink, Fugitive ink, Pen reactive ink, Heat reactive ink, Coin reactive inks, Migrating ink, Bleeding inks, Florescent ink, Metallic ink and UV ink..etc

Basic principles of security papers – Safety paper, Chemical reactive paper, Special papers, Water mark paper and Copy evident paper. .etc

Types and usages of security threads - Metalized thread, Windowed thread, Holographic windowed thread, Micro text, Clear text and Thermo text.

Basic principles of watermark, Classification of watermark - Line drawing watermark, shaded watermark and Digital watermark.

MODULE III

CURRENCY PRINTING

Creation & Graphics, Bank note security features , Applications of Computers in Designing.Currency- Signatures & numbering, Paper Specifications.

Watermark & Other Protective devices, Digital Watermark on Currency, Circulation & Bank

Maintenance. special issues counterfeiting.

CHEQUE PRINTING, NUMBERING AND BAR CODING

Introduction, Pre-Encoding and Post Encoding, Printing Tolerances, Testing equipment, E13B Characters, RBI Specifications, Principles of Cheque Design. Numbering with MICR Ink on Rotary presses, Trouble Shooting, Modulus Systems, Weighted & Unweighted.

Barcode: Introduction, types- Linear and matrix, barcode structure and characters, Principles of Bar Coding, Types of Coding- EAN 13 Code, Code 39 ACA etc., Typical Bar Code Machines & Print wheels, Scanners(barcode readers) and their functions. Applications of barcode.

MODULE IV

HOLOGRAMS, CREDIT CARDS & PASSPORTS

Hologram: Introduction, Manufacturing Process,. Holographic Recording & Master Origination. Finishing Process, Types of Holograms -reflection, transmission and hybrid, Security holograms, clickograms, stereogram, Anigram etc.

Credit card: introduction, security features, Materials used of specifications, Embossing, Magnetic Strip, Recording and Specifications, Manufacturing techniques. Security features in passports.

Reference :		
Author	Title	Publishers
Martin Monestics	The art of paper currency	Quarlet books ltd
Leibanger	Numbering machines and systems	Leibinger numbering systems
William H Erdei	Barcodes – design, printing and quality control	McGraw Hill